Customer Churn Analysis Report

End-to-End Analytics, Data Cleaning Workflow & Retention Strategy | Telecom

7,048	26.52%	1,869
TOTAL CUSTOMERS	OVERALL CHURN RATE	CHURNED ACCOUNTS

1. EXECUTIVE SUMMARY & OVERVIEW

This report presents a comprehensive end-to-end data analytics study of customer attrition (churn) within a telecommunications enterprise. By analyzing customer demographic profiles, subscribed services, account tenure, contract mechanics, and financial metrics across **7,048 customer accounts**, this project pinpoints primary drivers of customer defection and provides actionable, data-backed retention strategies to maximize customer lifetime value.

2. PROBLEM STATEMENT & ANALYTICAL SCOPE

Customer churn poses a significant threat to long-term profitability in telecommunications, as acquiring new customers costs up to five times more than retaining existing ones. The primary objectives of this analytical initiative are:

- Identify high-risk customer cohorts susceptible to early cancellation.
- Evaluate the impact of service types, payment methods, and contract terms on churn.
- Establish robust, automated Power Query data transformation standards.
- Deliver an interactive executive dashboard for real-time risk monitoring.

3. DATASET ARCHITECTURE

Dataset Structure & Key Variables (21 Attributes)

Demographics: Gender, Senior Citizen Status, Partner, Dependents.
Subscribed Services: Phone Service, Multiple Lines, Internet Service (DSL, Fiber Optic, No), Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies.
Account & Financial Mechanics: Tenure (Months), Contract Type (Month-to-month, One year, Two year), Paperless Billing, Payment Method, Monthly Charges, Total Charges, Churn Status (Yes/No).

4. DATA CLEANING & TRANSFORMATION WORKFLOW

Prior to visual modeling, raw transactional data (`telco_churn_unclean.csv`) was ingested and transformed using Power Query in Power BI Desktop to guarantee analytical accuracy:

Step 1 & 2: TotalCharges Blank & Error Handling
Identified blank and non-numeric entries in `TotalCharges` corresponding to new accounts with tenure ≤ 1 month. Converted column type to Decimal and applied *Replace Errors* to set empty values to 0 or null.

Step 3: MonthlyCharges Normalization
Removed currency symbol prefixes ($, ₹) across text strings using *Replace Values* and explicitly cast the column to Decimal Number.

Step 4: Contract Column Standardization
Standardized inconsistent text casing and hyphenation (e.g., unifying "month to month" and "Month-to-month" into a clean "Month-to-month" category).

Step 5: Whitespace Elimination (Trim & Clean)
Applied `Transform -> Format -> Trim` across all categorical text attributes (`PaymentMethod`, `InternetService`, `gender`, `Churn`) to remove trailing and leading spaces.

Step 6: Logical Error Correction & Conditional Mapping
Enforced strict business logic: if `PhoneService = 'No'`, set `MultipleLines = 'No'`. If `InternetService = 'No'`, set all add-on services (such as `StreamingTV`) to 'No'.

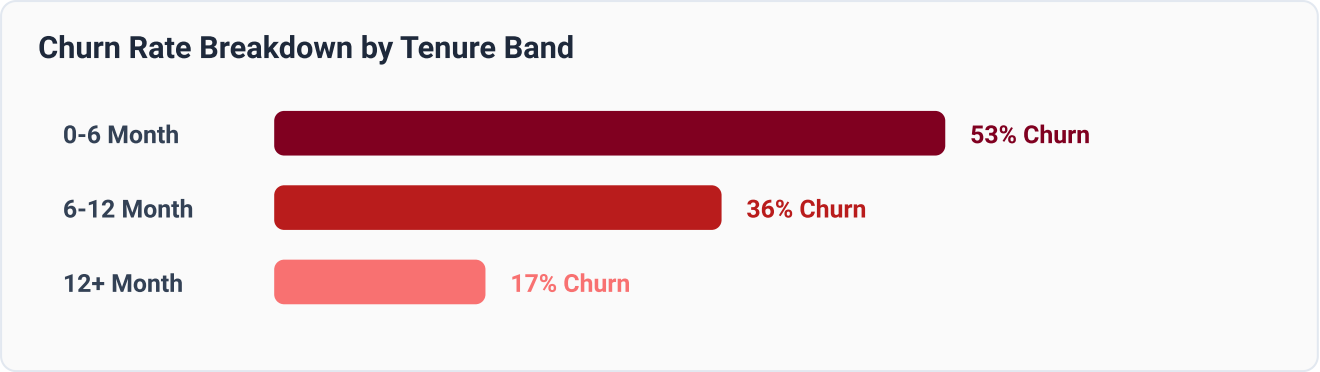
5. CORE FINDINGS & PRIMARY CHURN DRIVERS

Contract Vulnerability
Month-to-month contract holders account for over **88% of total churned customers**. Conversely, customers on 2-year contracts exhibit negligible churn (<3%).

High-Risk Retained Segment
Out of 5,179 retained customers, **633 active accounts** are classified as High Risk due to short tenure and non-contractual billing terms.

6. VISUAL ANALYTICS: TENURE BAND & CHURN IMPACT

A closer examination of customer tenure reveals that churn is heavily front-loaded during initial onboarding:



7. SERVICE TYPE BREAKDOWN & RISK PROFILE

Internet Service Type	Total Base	Churned Volume	Risk Category & Impact
Fiber Optic	3,096	1,297	High Risk (Represents 69.4% of total churn)
DSL	2,421	459	Moderate Risk
No Internet Service	1,531	113	Low Risk / Stable Base

8. STRATEGIC BUSINESS RECOMMENDATIONS

Retention & Growth Action Plan

1. **Onboarding & Early Success Interventions:** Deploy a dedicated 90-day onboarding program to guide new customers through setup, directly addressing the 53% early-stage churn rate.
2. **Contract Conversion Incentives:** Offer promotional add-on perks or targeted discounts to transition high-risk Month-to-Month customers into 1-Year or 2-Year agreements.
3. **Fiber Optic Service Quality & Pricing Audit:** Investigate network stability and pricing structures for Fiber Optic lines to resolve disproportionate defection.
4. **Proactive Outreach for High-Risk Accounts:** Leverage Power BI dashboard filters to identify and proactively contact the **633 high-risk retained accounts** before cancellation occurs.